Technopedia Arabia

Ahmed Darwish

Electrical Engineer | Embedded Systems & AI Engineer | Technical Trainer

CONTACT

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AT A GLANCE

16+ Years

150+ Projects

250+ Mentored

CORE SKILLS

Embedded / Arduino	95%
Circuit & PCB Design	90%
Python	88%
C / C++	85%
Simulation & SPICE	85%
AI & Computer Vision	82%
MATLAB / AutoCAD	78%
Training & Mentoring	92%

LANGUAGES

Arabic	Native
English	Very Good

Professional Summary

Electrical and embedded systems engineer with 20+ years spanning hands-on engineering and technical training. Specialist in Arduino, Raspberry Pi and microcontroller-based design, with strong Python and C++ for IoT, automation and computer vision. Delivered 150+ complete engineering projects and mentored 250+ engineers and students. My work bridges academic theory and real engineering: every circuit I teach is one I have designed, debugged and delivered for an actual client.

Experience

Embedded Systems Engineer & Circuit Designer
2010 – Present

Independent engineering practice — Kuwait & Worldwide

100+ Arduino and Raspberry Pi projects delivered across Kuwait and the region, with 40% more repeat business and 95% client satisfaction. Automation, IoT and smart-system solutions in Python and C++ cutting manual intervention by ~35%. End-to-end delivery: requirements, circuit and PCB design, prototyping, testing, handover. Trained 100+ engineers and students in hands-on workshops.

Lecturer, Electrical Engineering — Practical & Industrial Training
2015 – 2018 · 2025 – Present

Specialized Education & Training Centers — Kuwait

Undergraduate courses in circuit analysis, electronics, advanced computer applications and PC networking. Practical workshops in electrical control and contactors and bench fitting: hand tools, precision measurement, wire soldering, motor-control circuits, overload and earth-leakage protection, and fault troubleshooting under strict safety protocols. Supervised graduation projects from proposal to defense.

Curriculum Designer & Technical Educator — Electrical & Electronics
2009 – Present

Technical education sector — Kuwait

Designed and delivered Electrical Engineering and Electronics curricula, lifting engagement by ~30% and practical scores by ~25%. Labs and workshops integrating circuit design with embedded systems; digital tools and modern teaching methods introduced across 15+ classrooms. Mentored learners on Arduino, Raspberry Pi and sensor-based automation, contributing to a 20% rise in uptake of advanced technical tracks.

Executive Sales Engineer
2005 – 2008

Arab Engineers for Trading Co. — Saudi Arabia

Technical sales, bids, quotations, training and after-sales services for educational engineering equipment across Khobar, Dhahran and Jubail — including KFUPM, JIC and JTI. Pre-sales consultation and tailored engineering proposals.

Education & Certifications

B.Sc. Electronics & Electrical Communication Engineering 2004
Cairo University — Egypt

GSM Fundamentals & Survey 2005
HTC — Cairo University, Egypt

Sales of Specialized Educational Equipment 2006
PHYWE SYSTEME GmbH — Germany

- AI and Advanced Programming Languages — Specialized Seminar (2023)
- MS PowerPoint — Trained 150+ Participants (2022–2026)
- MS Word — Documentation techniques, +25% participant efficiency (2023–2025)

Software & Tools

PROGRAMMING
Python · C / C++ · MicroPython · JavaScript · Git / GitHub

EMBEDDED
Arduino IDE · PlatformIO · Raspberry Pi OS · ESP32 / ESP8266

CIRCUIT DESIGN & SIMULATION
Proteus · PSpice · Multisim · AutoCAD Electrical · PCB Layout

AI & COMPUTER VISION
TensorFlow · PyTorch · OpenCV · YOLOv5 · ONNX

ANALYSIS & DEPLOYMENT
MATLAB · Gradio · Hugging Face · MS Office

Key Achievements

- ✓ 150+ complete engineering projects, design to handover
- ✓ 250+ engineers and students trained in hands-on workshops
- ✓ ~30% lift in learner engagement, ~25% in practical scores
- ✓ 20% rise in uptake of advanced technical tracks
- ✓ ~35% less manual intervention via automation and IoT
- ✓ 95% client satisfaction, 40% increase in repeat business
- ✓ ~20% fewer design errors through simulation tooling
- ✓ Digital tools and modern methods across 15+ classrooms